A7 GROUP REAL ESTATE

Anwar Al Riyadh Properties • Dubai

WhatsApp Business Communication & Lead-Control System

Technical Specification & Developer Build Guide

Prepared for

The A7 Group Development Team (Lahore) — n8n / Backend / Mobile Engineers

Purpose: A complete, build-ready blueprint. No feature should be missed.

Target build time: 15 working days with 2–3 engineers using AI-assisted development.

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1. Executive Summary — What We Are Building and Why

This document describes a single, connected system that lets A7 Group agents talk to customers on WhatsApp using the company's own number — never their personal number — while the company keeps full control and visibility of every conversation.

Today, agents pull leads out of Bitrix24, message customers from their personal WhatsApp, and close deals privately. Management cannot see what was said, cannot protect the lead, and cannot stop an agent from taking the client with them if they leave. This system closes that gap completely.

The one-line goal

> **Goal**
> Every customer conversation happens on the COMPANY WhatsApp number, is visible to management in real time, is logged automatically into Bitrix24, and works from an agent's phone while they are out doing viewings — with no laptop required.

What the agent experiences

- **One app.** A simple app on their phone (like a normal chat app) that shows their assigned customer chats.
- **Chat normally.** They tap a customer, type, and send — the message goes out from the company WhatsApp number.
- **One-tap re-open.** If the 24-hour window is closed, one button sends an approved template to re-open the chat.
- **No personal number.** They never see or use their personal WhatsApp for company customers.

What the manager (you) experiences

- **See everything.** A live dashboard showing every agent's conversations as they happen.
- **Smart alerts.** Automatic flags: serious buyer, competitor agent, duplicate client, poor agent behaviour.
- **One source of truth.** Everything already sitting inside / linked to Bitrix24 — no double data entry.

2. The Business Problems This Solves

Each feature in this build exists to solve a specific, real problem A7 Group faces every day. Engineers should keep these in mind — the goal is the business outcome, not just the code.

Problem today	How this system fixes it
Agents talk to customers on personal WhatsApp; management is blind to it.	All chats run through the company number and are visible + logged automatically.
Leads get 'stolen' — an agent leaves and takes the client.	The company owns the number and the chat history. The relationship stays with A7 Group.
Agents are out on viewings with only a phone, not a laptop.	A mobile app lets them do everything from the phone, in the field.
After 24 hours WhatsApp blocks free messages; agents give up and use personal WhatsApp.	One-tap approved templates re-open the chat instantly. No friction.
Two or three agents unknowingly chase the same client and undercut each other.	Duplicate-client detection alerts management the moment it happens.
Competitor agents pose as buyers to extract owner details and cut A7 out.	A 30k–40k agent database auto-flags them before the agent even replies.
No idea which leads are hot.	AI reads each chat and scores how serious the customer is.
Template/utility messages cost money if sent carelessly.	The system uses the free 24-hour window first and only sends paid templates when needed.

3. System Architecture (The Big Picture)

The system has five moving parts. Everything is glued together by one central Backend Server that A7 Group owns and controls.

The five parts

- **1) WhatsApp Cloud API.** Meta's WhatsApp Business Cloud API — already owned by A7 Group. Sends and receives the actual WhatsApp messages.
- **2) Backend Server.** A Node.js server that receives every incoming message (webhook), stores it, applies the AI + detection logic, and pushes it to the app, dashboard and Bitrix24. This is the brain.
- **3) Agent Mobile App.** The phone app agents use in the field. It talks ONLY to the Backend Server, never directly to Meta.
- **4) Manager Dashboard.** A web dashboard for you and sales managers to watch and control everything.
- **5) Bitrix24 + n8n.** Already owned. The backend writes contacts, deals and chat history into Bitrix24 so the CRM stays the record of truth. n8n handles the automation glue.

How a message travels (incoming)

1. Customer sends a WhatsApp message to the company number.
2. Meta forwards it to the Backend Server via webhook.
3. Backend saves the message, checks: is this number a known competitor agent? Is this client already talking to another agent? What is the AI sentiment?
4. Backend pushes the message + flags to the assigned agent's app and to the Manager Dashboard in real time.
5. Backend writes/updates the contact and chat in Bitrix24.

How a message travels (outgoing)

6. Agent types a reply in the app (or taps a template button).
7. App sends it to the Backend Server.
8. Backend checks the 24-hour window: if open, send free-form; if closed, send an approved template.
9. Backend calls Meta's WhatsApp API to deliver the message from the company number.
10. Backend logs the outgoing message to the database, dashboard and Bitrix24.

> **Key design rule**
> The Agent App NEVER connects directly to Meta or holds any WhatsApp API key. It only talks to our Backend Server. This is what keeps the company in full control and stops any agent from taking the number or the data with them.

Recommended technology

Layer	Recommended tech	Why
Backend server	Node.js + Express (or NestJS)	Fast, huge talent pool, works perfectly with WhatsApp Cloud API & Bitrix24 REST.
Database	PostgreSQL	Reliable, handles the message + agent-database volume easily.
Realtime	Socket.IO or WebSocket	Pushes new messages to app & dashboard instantly.

Layer	Recommended tech	Why
Mobile app	React Native	One codebase for Android + iPhone; agents install on any phone.
Dashboard	React + Tailwind	Fast to build, clean, works on any browser.
AI sentiment	OpenAI / Claude API	Reads chats and scores lead seriousness with no training needed.
Automation glue	n8n (already owned)	Connects backend events to Bitrix24 actions without heavy coding.
Hosting	AWS (Lightsail/EC2 — already used)	A7 Group already runs Bitrix24 here; keep it together.

4. Component 1 — WhatsApp Business API Layer

A7 Group already has WhatsApp Business API access. This section tells the engineer exactly how to wire it into the Backend Server.

4.1 What must be in place

- A verified Meta Business account and a WhatsApp Business API phone number (already owned).
- A permanent System User access token (not a temporary token) so the connection never expires.
- A registered Webhook URL pointing to our Backend Server, subscribed to 'messages' events.
- Pre-approved message templates in the Meta Business Manager (see Section 6).

4.2 Receiving messages (Webhook)

The Backend exposes one public HTTPS endpoint, e.g. /webhook/whatsapp. Meta sends a GET request once to verify it (echo the challenge token), then POSTs every incoming message to it.

For each incoming message the backend must capture: sender phone number, message text/media, timestamp, and the WhatsApp message ID. Store it, then run the detection pipeline (Sections 8, 10, 11).

4.3 Sending messages

The backend sends messages by calling Meta's Cloud API endpoint /{phone-number-id}/messages with the company token. Two message types matter:

- **Session message.** Free-form text — allowed only inside the open 24-hour window.
- **Template message.** Pre-approved template — required to start a chat or re-open a chat after 24 hours.

> **Security note for the engineer**
> The WhatsApp token lives ONLY on the Backend Server, stored as an environment variable / secret. It is never shipped inside the mobile app or the dashboard code. If it ever leaks, rotate it immediately in Meta Business Manager.

5. Component 2 — Agent Mobile App (React Native)

This is the app the agent installs on their phone and uses in the field. It must feel as simple as a normal chat app so agents actually use it and never fall back to personal WhatsApp.

5.1 Screens the app must have

Screen	What it does
Login	Agent logs in with company credentials. No WhatsApp key is stored on the phone.
Chat list	All customers assigned to this agent, newest first, with unread badges and flags (🔥 serious, ⚠ competitor, 👥 duplicate).
Chat window	The full conversation. Type & send. Shows if the 24-hour window is open or closed.
Template button	When window is closed, a one-tap menu of approved templates (property list, follow-up, greeting, etc.).
Customer card	Name, community interest, budget, Bitrix24 link, and any auto-flags on this contact.
Notifications	Push alerts for new customer messages even when the app is closed.

5.2 Rules built into the app

- **Least access.** An agent sees ONLY their own assigned chats — not the whole company's leads.
- **Company number only.** Sending always goes through the Backend, which stamps the company number as sender.
- **No dead ends.** If the 24-hour window is closed, the free-text box is disabled and the template menu is shown instead — the agent physically cannot get stuck.
- **Full audit.** Every action (open chat, send, send template) is logged with agent ID and time.

5.3 Why an app and not just a website

Malik's requirement: when an agent is out on a viewing they must NOT have to open a browser and log into a website. A native app gives push notifications, one-tap open from the home screen, and works smoothly on a phone in the field. The app is the priority; a browser version of the dashboard is only for managers at a desk.

6. Component 3 — The 24-Hour Window & Template System

This is the single most important feature for adoption. If agents find it hard to message a customer after a day or two, they will go back to personal WhatsApp. This must be effortless.

6.1 How WhatsApp's rule works

- When a customer messages you, a 24-hour 'service window' opens. Inside it you can send any free-form message for free.
- After 24 hours of customer silence, the window closes. You can then only reach them with a pre-approved TEMPLATE message.
- A7 Group's reality: customers often go quiet for 2–3 days while the team finds the right property, then the agent needs to re-open the chat. This is normal and must be one tap.

6.2 What the system does automatically

11. The backend tracks, per customer, the time of their last inbound message.
12. The app shows a clear status: green 'Open — reply freely' or grey 'Closed — send a template to re-open'.
13. When closed, the agent taps one template from a ready list; the backend sends the approved Meta template with the customer's name filled in.
14. The customer replies → the 24-hour window re-opens → the agent chats freely again.

6.3 Templates to pre-approve in Meta (starting set)

Template name	Purpose	Category
welcome_intro	First contact / greeting a new lead.	Marketing / Utility
property_ready	'We found properties matching your needs' re-open.	Utility
follow_up	Gentle follow-up after silence.	Utility
viewing_confirm	Confirm a viewing date/time.	Utility
document_request	Ask for documents (Emirates ID, etc.).	Utility

Each template should have 2–3 rotating versions (worded differently) to stay compliant with anti-spam rules — the same pattern already used in A7 Group's owner-outreach packs.

Adoption is the whole game
If this feature takes more than ONE tap, agents will abandon the system. Build the template menu to be instant, pre-filled with the customer's name, and impossible to get wrong.

7. Component 4 — Bitrix24 Two-Way Sync

Bitrix24 stays the company's record of truth. The agent never has to enter anything twice — the backend keeps Bitrix24 updated automatically.

7.1 What syncs into Bitrix24

- **Contacts.** Every new customer becomes / updates a Contact in Bitrix24.
- **Chat history.** Every conversation is attached to that contact (as timeline/chat history or a linked deal).
- **Flags & scores.** AI sentiment score, competitor-agent flag, and duplicate flag are written to custom fields on the contact.
- **Deals.** Serious leads can auto-create or move a Deal in the sales pipeline.

7.2 How it connects

The backend uses the Bitrix24 REST API (inbound webhook) already available on A7 Group's on-premise Bitrix24. n8n can handle much of this glue: backend fires an event → n8n workflow → Bitrix24 action. This keeps heavy CRM logic out of the core code and easy to change later.

Custom fields to add in Bitrix24: Sentiment Score, Lead Status (Hot/Warm/Cold), Is Competitor Agent (Y/N), Duplicate Alert (Y/N), Assigned Agent, Last WhatsApp Contact.

8. Component 5 — AI Sentiment & Serious-Lead Detection

AI reads each conversation and tells you which customers are serious, so agents and managers focus energy where it matters.

8.1 What the AI decides

Signal	Meaning
Sentiment (positive / neutral / negative)	The customer's mood and interest level.
Seriousness score (0–100)	How likely this is a real, ready buyer/tenant vs a time-waster.
Lead status (Hot / Warm / Cold)	A simple label managers can act on instantly.
Intent	Buying, renting, just browsing, or (importantly) a competitor agent fishing.

8.2 How it works

15. After each customer message (or every few messages), the backend sends the recent conversation to the AI API with a fixed instruction: 'Score this real-estate lead's seriousness 0–100, give sentiment, status and intent, reply in JSON only.'
16. The backend saves the scores and updates the app, dashboard and Bitrix24.
17. If the score crosses a 'hot' threshold, the manager gets an instant alert: 'Serious customer — Agent X, Community Y.'

> **Cost tip**
> Do not call the AI on every single word. Batch it — score after a customer replies, or every 3–4 messages. This keeps AI costs low while still catching hot leads fast.

9. Component 6 — Manager Dashboard

The web dashboard is where you, the sales manager and the CEO watch the whole operation live. This is your control tower.

9.1 What the dashboard shows

- **Live chats.** Every agent's live conversations, updating in real time.
- **Hot leads.** A 'Hot Leads' panel — all serious customers right now, across all agents.
- **Alerts.** Alerts feed: competitor agent detected, duplicate client detected, agent misbehaviour.
- **Agent scoreboard.** Per-agent stats: messages sent, response time, hot leads handled, deals moved.
- **Search.** Search any customer / number and see the full history instantly.

9.2 Roles

Role	Can see / do
Agent	Only their own assigned chats (in the app).
Sales Manager	All chats of their team + alerts + stats.
CEO / Managing Director (you)	Everything, all teams, plus admin controls.

10. Component 7 — Duplicate-Client Detection

Sometimes one customer contacts two or three A7 Group agents. The agents then compete against each other, undercut on price, and the deal breaks or margin is lost. The company must know instantly.

10.1 The logic

18. Every incoming customer number is checked against all currently active conversations.
19. If the same number is already open with a different agent, the system raises a Duplicate-Client alert.
20. The alert goes to both agents' managers and appears on the dashboard: 'Client +9715XXXX is talking to Agent A and Agent B.'
21. Management decides who keeps the client. The system can then re-assign the chat to one agent.

Business value

This single feature stops internal price wars and 'deal break' situations that cost A7 Group money and reputation. One customer, one agent, decided by management — not by whoever shouts loudest.

11. Component 8 — Competitor-Agent Detection

Many callers on Property Finder and Bayut are not real buyers — they are competitor agents fishing for property details, who then go directly to the owner to avoid paying A7 Group a commission. A7 Group already has a database of 30,000–100,000 agent numbers. The system uses it to unmask them automatically.

11.1 The logic

22. A7 Group's agent-number database (30k–100k numbers) is loaded into a dedicated, indexed table.
23. The moment any call or WhatsApp arrives, the number is checked against this list — before the agent even replies.
24. If it matches, the app and dashboard show a clear banner: 'THIS IS A COMPETITOR AGENT.'
25. The agent instantly knows to change approach — share less, protect the owner's details, and follow company policy on what not to reveal.

11.2 Keeping the list fresh

- Admin can bulk-upload new agent numbers (Excel/CSV) any time.
- If a flagged 'agent' turns out to be a real buyer, admin can remove or whitelist them.
- The system can suggest new agent numbers to add when a caller behaves like an agent (optional AI assist).

> **Compliance reminder**
> This database is used defensively — to recognise who is calling YOU and protect owner details. It must be used in line with UAE data-protection rules (PDPL) and TDRA telemarketing regulations. Do not use these numbers for unsolicited cold marketing.

12. Component 9 — Quality Monitoring & Admin Controls

12.1 Quality monitoring

- **Behaviour flags.** AI flags rude, unprofessional, or off-policy messages by agents.
- **Response time.** Slow responses (customer waiting too long) are highlighted.
- **Coaching.** Managers can review any conversation and leave a private coaching note.

12.2 Admin controls (for you)

- Add / remove agents and assign them to communities.
- Add / edit message templates and their rotating versions.
- Upload / update the competitor-agent database.
- Set the 'hot lead' threshold and alert rules.
- Re-assign a customer from one agent to another in one click.
- Export any conversation or report.

13. Database Schema

A simple, clean PostgreSQL schema. This is enough to build the whole system; engineers can add indexes and fields as needed.

agents

Field	Type	Notes
id	UUID	Primary key
name	text	Agent full name
phone	text	Agent's own phone (for login only)
role	text	agent / manager / admin
community	text	Assigned community/pod
active	boolean	Enabled or disabled

customers

Field	Type	Notes
id	UUID	Primary key
wa_number	text	Customer WhatsApp number (E.164)
name	text	From WhatsApp profile / manual
assigned_agent_id	UUID	FK → agents.id
bitrix_contact_id	text	Linked Bitrix24 contact
sentiment_score	int	0–100 latest AI score
lead_status	text	hot / warm / cold
is_competitor_agent	boolean	Matched against agent DB
duplicate_flag	boolean	Talking to >1 agent
last_inbound_at	timestamp	Drives the 24-hour window

messages

Field	Type	Notes
id	UUID	Primary key
customer_id	UUID	FK → customers.id
agent_id	UUID	FK → agents.id (null if inbound)
direction	text	inbound / outbound
body	text	Message content
is_template	boolean	Was a template used
wa_message_id	text	Meta's message ID
created_at	timestamp	Time

competitor_agents

Field	Type	Notes
id	UUID	Primary key
phone	text	Agent number (indexed for fast lookup)
source	text	Where the number came from
added_at	timestamp	When added

templates

Field	Type	Notes
id	UUID	Primary key
name	text	Meta template name
versions	jsonb	Rotating worded versions
category	text	utility / marketing

14. WhatsApp Message Cost Control (Meta Pricing)

Meta charges for WhatsApp messages based on message category and whether you are inside the free service window. The system is designed to minimise cost automatically.

14.1 The key facts

- **Free window.** Messages a customer sends to you, and your replies inside the 24-hour window, are effectively free (service conversations).
- **Utility < Marketing.** Utility templates (order/appointment/follow-up style) are cheaper than Marketing templates.
- **Re-open cost.** Re-opening a chat after 24 hours needs a template, which has a cost — so the system uses the free window whenever possible.

> **Important — verify current pricing**
>
> Meta changes WhatsApp pricing periodically (and moved toward per-message pricing for utility/marketing categories). Before go-live, the engineer must confirm the CURRENT UAE rates in the Meta Business Manager, because exact per-message fees change over time.

14.2 How the system saves money

26. Always prefer a free-form reply inside the open 24-hour window.
27. When a template is required, default to the cheaper Utility category, not Marketing, wherever the message qualifies.
28. Show the manager a monthly report: number of free messages vs paid templates, and estimated spend — so cost is always visible.

15. 15-Day Implementation Plan (Day by Day)

This plan assumes 2–3 engineers working in parallel, using AI coding tools (GitHub Copilot / ChatGPT / Claude) to move fast. A7 Group already owns the WhatsApp API, Bitrix24 and n8n, which removes the slowest setup steps.

Team split

- **Engineer 1:** Backend + WhatsApp API + Bitrix24 + detection logic.
- **Engineer 2:** React Native mobile app.
- **Engineer 3:** Manager dashboard + AI sentiment + n8n glue (can be shared with Engineer 1 if only 2 people).

Day-by-day

Days	Work	Owner
1–2	Set up server, database, WhatsApp webhook receiving & sending working end-to-end.	Eng 1
1–2	Mobile app skeleton: login + chat list + chat window (dummy data).	Eng 2
3–4	Connect app to backend: real messages send & receive from company number.	Eng 1+2
3–4	24-hour window logic + template send (one-tap re-open).	Eng 1
5–6	Bitrix24 sync: contacts, chat history, custom fields via REST/n8n.	Eng 3
5–7	Manager dashboard: live chats, agent list, search.	Eng 3
7–8	AI sentiment scoring + hot-lead alerts.	Eng 3
8–9	Competitor-agent database load + real-time lookup + banner.	Eng 1
9–10	Duplicate-client detection + alerts.	Eng 1
10–11	Quality monitoring, admin panel (templates, agents, thresholds).	Eng 3
11–12	Push notifications, polish app UX, role permissions.	Eng 2
12–13	Full integration testing (see Section 16).	All
13–14	Fix bugs, load real competitor database, real templates approved in Meta.	All
14–15	Deploy to production, train a pilot group of agents, go live.	All

Realistic expectation

15 days is achievable for a working system with 2–3 focused engineers and AI assistance, BECAUSE the WhatsApp API, Bitrix24 and n8n already exist. If only one engineer is available, expect 26–32 days. Build the mobile app + WhatsApp + 24-hour window FIRST (that is what agents touch daily); dashboards and AI can follow in week 2.

16. Testing Checklist

Do not go live until every box below passes.

- ☐ Customer sends a WhatsApp message → it appears in the assigned agent's app within seconds.
- ☐ Agent replies in the app → customer receives it from the COMPANY number (not the agent's personal number).
- ☐ Message appears in the Manager Dashboard live.
- ☐ Message and contact appear in Bitrix24.
- ☐ After 24 hours of silence, the free-text box is disabled and the template menu appears.
- ☐ One tap on a template re-opens the chat successfully.
- ☐ AI sentiment score appears and updates as the chat continues.
- ☐ A hot lead triggers a manager alert.
- ☐ A known competitor-agent number triggers the 'Competitor Agent' banner instantly.
- ☐ The same customer talking to two agents triggers a Duplicate-Client alert.
- ☐ Agent can only see their own chats; manager sees the whole team.
- ☐ Push notification arrives when the app is closed.
- ☐ Admin can upload competitor numbers, add templates, and re-assign a customer.
- ☐ Cost report shows free vs paid messages for the month.
- ☐ No WhatsApp API key is present anywhere in the mobile app or dashboard code.

17. Deployment & Go-Live Guide

29. Host the backend on A7 Group's AWS (same environment as Bitrix24) with HTTPS.
30. Store all secrets (WhatsApp token, Bitrix24 webhook, AI key) as server environment variables — never in code.
31. Register the production webhook URL in Meta Business Manager.
32. Get all message templates approved in Meta (allow 1–2 days for Meta review).
33. Load the real competitor-agent database (Excel → import).
34. Create agent accounts and assign communities.
35. Publish the mobile app: Android APK / Google Play internal, and iOS via TestFlight or App Store.
36. Run a pilot with 3–5 agents for one week before rolling out to the full team.
37. Once stable, mandate the app company-wide and disable personal-WhatsApp use for company customers via the signed agent policy.

18. What to Give the Engineers on Day 1

Hand the team this list so nothing blocks the start.

- Meta Business account access + WhatsApp Business API phone number ID and permanent token.
- Bitrix24 inbound webhook URL and API access.
- Access to the AWS environment (or a new server) for hosting.
- n8n access (already owned).
- An AI API key (OpenAI or Claude) for sentiment analysis.
- The competitor-agent Excel database (30k–100k numbers).
- A7 Group logo + brand colours (navy & gold) for the app and dashboard.
- The list of communities and which agents belong to each.
- This specification document.

A7 Group Real Estate — Confidential

This blueprint is designed so no requirement is missed. Build the daily-use pieces first, then layer intelligence on top.